GDP Growth and Unemployment in the Netherlands and Western Europe (2015–2025)

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```r
library(tidyverse)
library(maps)

# Load data
gdp_data           <- read_csv("data/raw_gdp.csv")
unemployment_sex   <- read_csv("data/raw_unemployment_sex.csv")
unemployment_youth <- read_csv("data/raw_unemployment_youth.csv")

# Clean data
gdp_clean <- gdp_data %>%
  filter(geo == "Netherlands") %>%
  select(TIME_PERIOD, OBS_VALUE) %>%
  rename(Year = TIME_PERIOD, GDP_Growth = OBS_VALUE) %>%
  filter(!is.na(GDP_Growth))

unemp_clean <- unemployment_sex %>%
  filter(geo == "Netherlands") %>%
  select(TIME_PERIOD, OBS_VALUE) %>%
  rename(Year = TIME_PERIOD, Unemp_Rate = OBS_VALUE) %>%
  mutate(Unemp_Rate = ifelse(Unemp_Rate == -99, NA, Unemp_Rate)) %>%
  filter(!is.na(Unemp_Rate))

youth_clean <- unemployment_youth %>%
  filter(geo == "Netherlands") %>%
  select(TIME_PERIOD, OBS_VALUE) %>%
  rename(Year = TIME_PERIOD, Youth_Unemp_Rate = OBS_VALUE) %>%
  mutate(Youth_Unemp_Rate = ifelse(Youth_Unemp_Rate == -99, NA, Youth_Unemp_Rate)) %>%
  filter(!is.na(Youth_Unemp_Rate))

# Merge and engineer variables
combined_data <- gdp_clean %>%
  left_join(unemp_clean, by = "Year") %>%
  left_join(youth_clean, by = "Year") %>%
  mutate(Youth_Vulnerability_Ratio = Youth_Unemp_Rate / Unemp_Rate)
```

Set-up your environment

GDP Growth and Unemployment in the Netherlands and Western Europe

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Part 1 - Identify a Social Problem

1.1 Describe the Social Problem

Unemployment is one of the most closely watched indicators of economic well-being, because joblessness carries costs that go beyond lost income: it is associated with poorer mental and physical health, lower lifetime earnings, and weaker social cohesion (Eurofound, 2017). A central question for policymakers is how unemployment responds to the wider economy. Okun's Law describes a stable negative relationship between output growth and unemployment (Okun, 1962): when GDP grows, unemployment tends to fall, and vice versa.

Understanding the strength of this link matters for policy. If unemployment reacts sharply to downturns, governments may need stronger automatic stabilisers; if it reacts weakly — as during the 2020 COVID-19 shock, when wage-subsidy schemes kept workers employed — the usual relationship can break down. This report quantifies the relationship between GDP growth and unemployment in the Netherlands (2015-2025), compares unemployment across Western Europe, and examines how youth workers are disproportionately exposed.

Part 2 - Data Sourcing

2.1 Load in the data

We use three datasets downloaded from Eurostat, the statistical office of the European Union. The exact tables are:

- Real GDP growth rate: Eurostat table `tec00115` (annual % change).
- Total unemployment rate: Eurostat table `tesem120`.
- Youth unemployment rate (ages 15-24): Eurostat table `tipslm80`.

The raw extracts are stored in the `data/` folder and loaded in the pipeline at the top of this document.

2.2 Provide a short summary of the dataset(s)

Each dataset reports an annual percentage value per country. The unemployment series express the number of unemployed as a share of the active labour force; the GDP series expresses the annual percentage change in real GDP. All values are sourced from Eurostat and cover the period 2015-2025.

2.3 Describe the type of variables included

Think of things like:

- Do the variables contain health information or SES information?
- Have they been measured by interviewing individuals or is the data coming from administrative sources?

The variables are socio-economic indicators. Unemployment figures come from the EU Labour Force Survey, a large harmonised household survey, while GDP figures come from national accounts compiled by national statistical institutes. The data is therefore official administrative/survey data, comparable across countries, rather than self-reported by individuals.

Part 3 - Quantifying

3.1 Data cleaning

All cleaning steps were run in the data pipeline at the top of this file to keep the workflow reproducible. For each dataset we (1) filtered for the Netherlands, (2) kept only the year and value columns, (3) recoded the placeholder -99 as NA and dropped missing values, and (4) joined the three series by year.

3.2 Generate necessary variables

Variable 1: Youth Vulnerability Ratio

This variable divides the youth unemployment rate by the total unemployment rate (`Youth_Unemp_Rate / Unemp_Rate`), created in the pipeline at the top. It shows how much more exposed young workers are than the workforce as a whole.

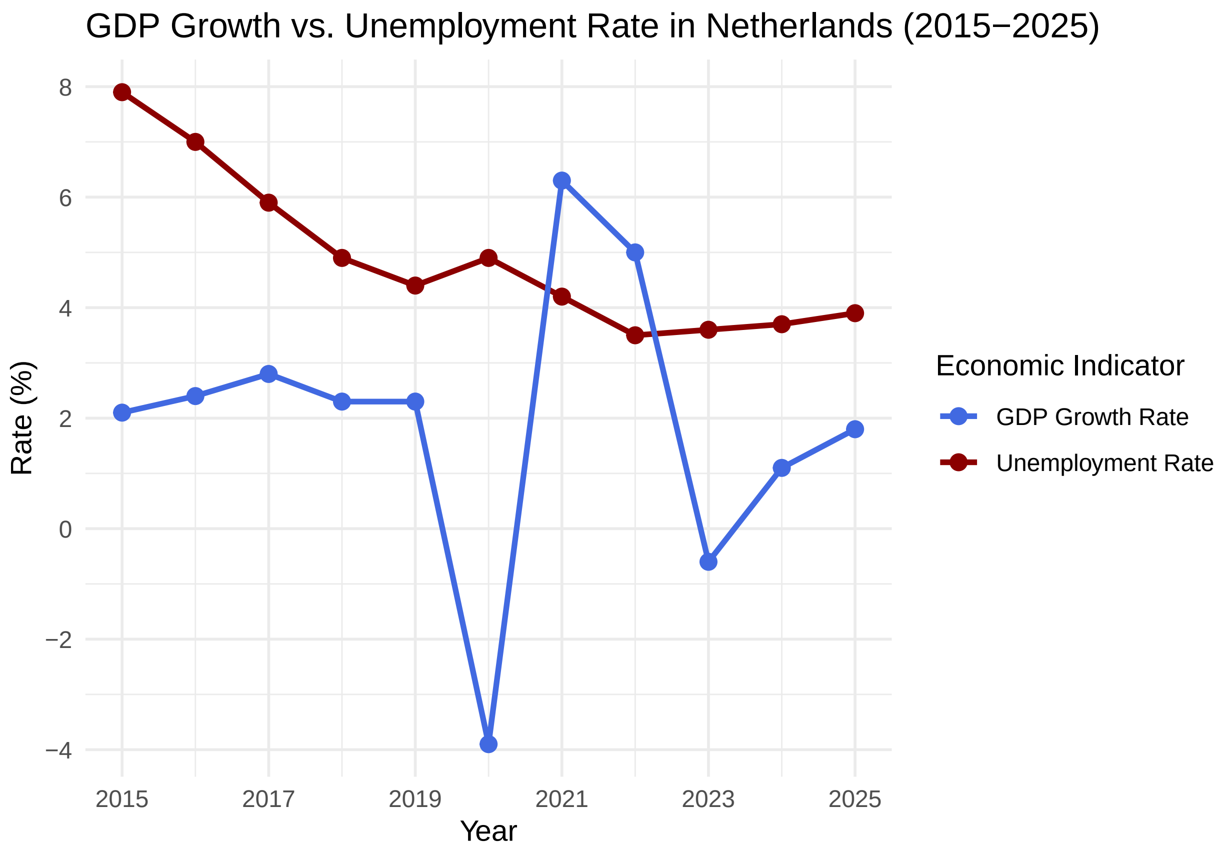
Variable 2: Annual change in unemployment (percentage points)

We construct the year-on-year change in the unemployment rate (`Ppt_Change`) to capture labour-market dynamics rather than levels. It is created in the event-analysis chunk in 3.6 using `Unemp_Rate - lag(Unemp_Rate)`, and is used to quantify how sharply unemployment moved around the 2020 shock.

3.3 Visualize temporal variation

```r
ggplot(data = combined_data, aes(x = Year)) +
  geom_line(aes(y = Unemp_Rate, color = "Unemployment Rate"), linewidth = 1) +
  geom_point(aes(y = Unemp_Rate, color = "Unemployment Rate"), size = 2.5) +

  geom_line(aes(y = GDP_Growth, color = "GDP Growth Rate"), linewidth = 1) +
  geom_point(aes(y = GDP_Growth, color = "GDP Growth Rate"), size = 2.5) +

  scale_color_manual(values = c("Unemployment Rate" = "darkred", "GDP Growth Rate" = "royalblue")) +
  scale_x_continuous(breaks = seq(2015, 2025, by = 2)) +
  scale_y_continuous(breaks = seq(-4, 10, by = 2)) +
  theme_minimal() +
  labs(
    title = "GDP Growth vs. Unemployment Rate in Netherlands (2015-2025)",
    x = "Year",
```

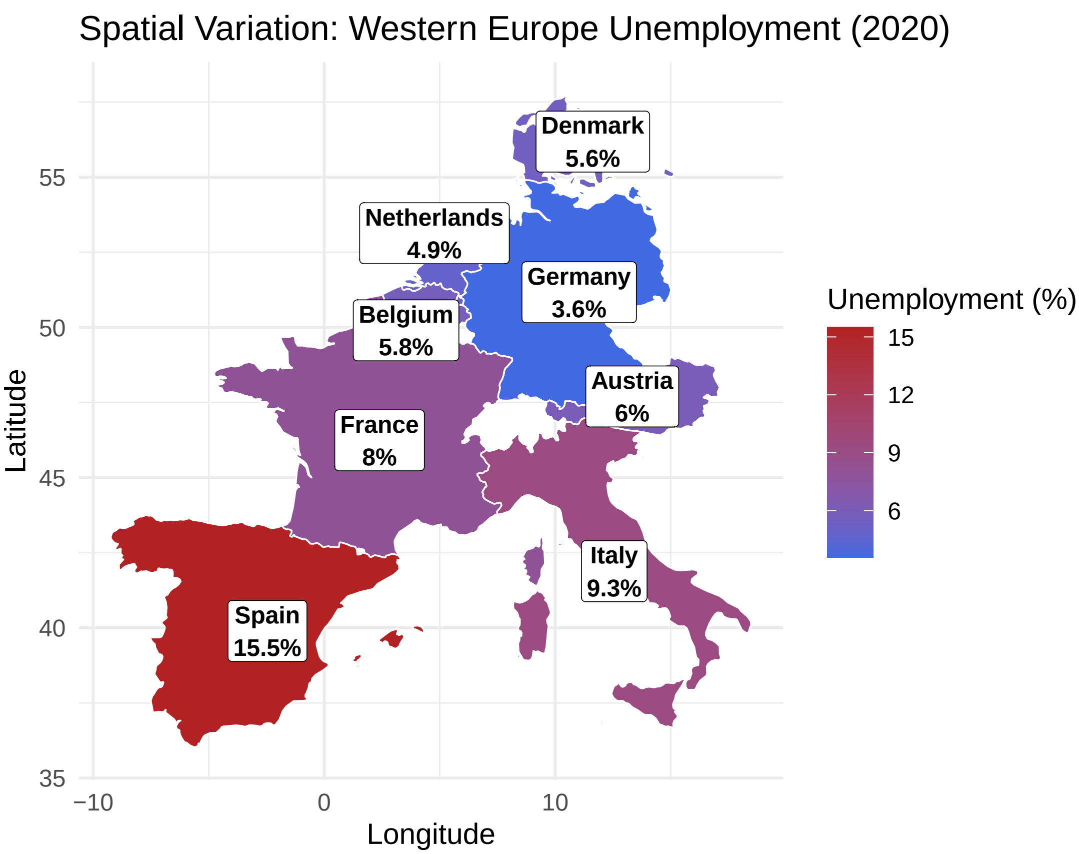
```r
    y = "Rate (%)",
    color = "Economic Indicator"
  )
```



3.4 Visualize spatial variation

```r
europe_map <- map_data("world") %>%
  filter(region %in% c("Netherlands", "Germany", "Belgium", "France", "Italy", "Spain", "Austria", "Deni

europe_unemp_2020 <- unemployment_sex %>%
  filter(TIME_PERIOD == 2020) %>%
  rename(region = geo)

map_data_joined <- left_join(europe_map, europe_unemp_2020, by = "region")

country_labels <- map_data_joined %>%
  group_by(region) %>%
  summarise(
    long = mean(range(long)),
    lat = mean(range(lat)),
    Unemp_Rate = first(OBS_VALUE)
  ) %>%
  mutate(
```

```
    lat = case_when(
      region == "Netherlands" ~ lat + 1.0,
      region == "Belgium" ~ lat - 0.6,
      TRUE ~ lat
    ),
    long = case_when(
      region == "Netherlands" ~ long - 0.5,
      region == "Belgium" ~ long - 0.9,
      region == "Germany" ~ long + 0.6,
      TRUE ~ long
    )
  )

ggplot(map_data_joined, aes(x = long, y = lat, group = group, fill = OBS_VALUE)) +
  geom_polygon(color = "white", linewidth = 0.3) +
  scale_fill_gradient(low = "royalblue", high = "firebrick", na.value = "gray80", name = "Unemployment

  geom_label(data = country_labels,
             aes(x = long, y = lat, label = paste0(region, "\n", Unemp_Rate, "%")),
             inherit.aes = FALSE,
             fill = "white", color = "black", size = 3.0, fontface = "bold",
             label.padding = unit(0.2, "lines"), label.size = 0.15) +

  coord_fixed(1.3) +
  theme_minimal() +
  labs(title = "Spatial Variation: Western Europe Unemployment (2020)", x = "Longitude", y = "Latitude")
```

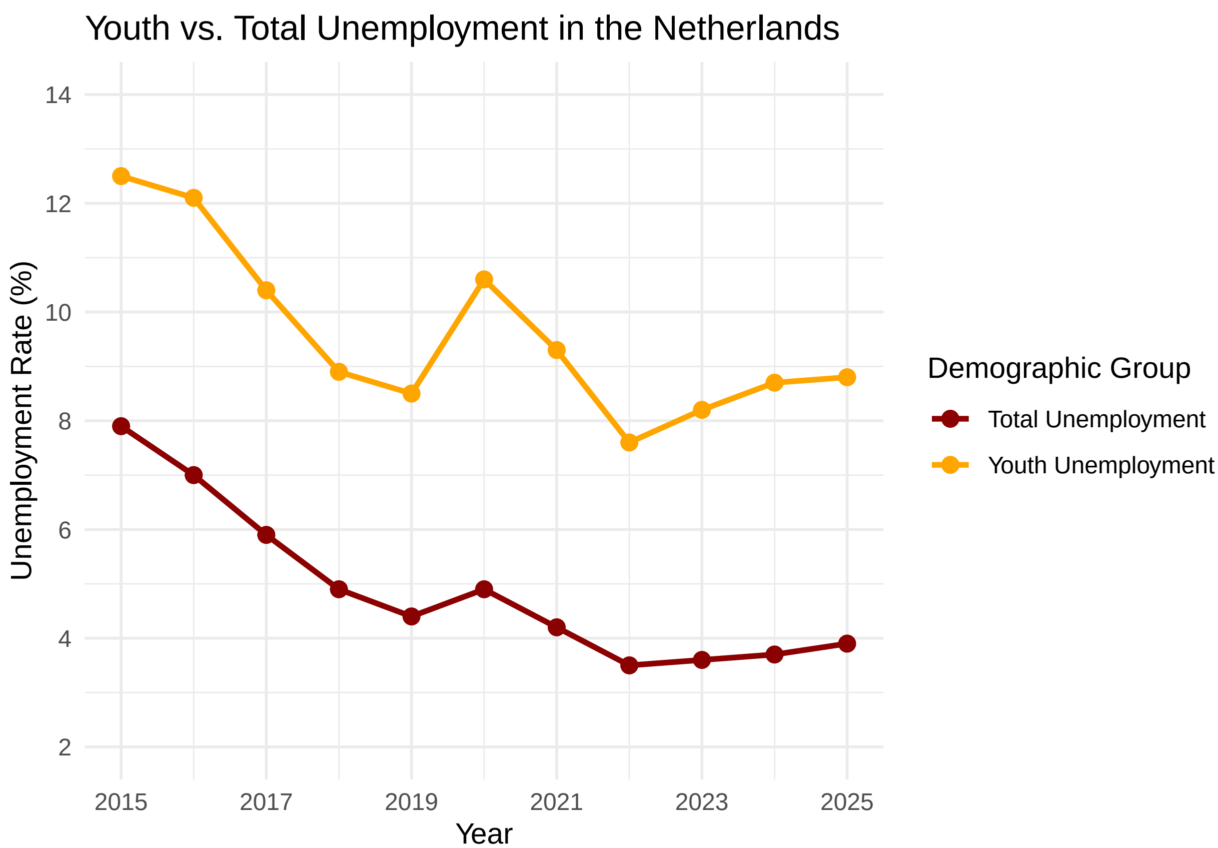


The map shows wide spatial variation in 2020 unemployment across Western Europe. The Netherlands (4.9%) and Germany (3.6%) sit at the low end, while Spain reaches 15.5% and Italy 9.3%. This clear north-south gradient places the Dutch labour market in context as relatively resilient, and motivates focusing the time-series analysis on the Netherlands.

3.5 Visualize sub-population variation

```
df_total <- unemp_clean %>%
  rename(Unemployment_Rate = 2) %>%
  mutate(Demographic_Group = "Total Unemployment")

df_youth <- youth_clean %>%
  filter(Year >= 2015 & Year <= 2025) %>%
  rename(Unemployment_Rate = 2) %>%
  mutate(Demographic_Group = "Youth Unemployment")

youth_data <- bind_rows(df_total, df_youth)

ggplot(data = youth_data, aes(x = Year, y = Unemployment_Rate, color = Demographic_Group, group = Demog
  geom_line(linewidth = 1) +
  geom_point(size = 2.5) +
  scale_color_manual(values = c("Total Unemployment" = "darkred", "Youth Unemployment" = "orange")) +
  scale_x_continuous(breaks = seq(2015, 2025, by = 2)) +
  scale_y_continuous(limits = c(2, 14), breaks = seq(2, 14, by = 2)) +
```

```r
  theme_minimal() +
  labs(
    title = "Youth vs. Total Unemployment in the Netherlands",
    x = "Year",
    y = "Unemployment Rate (%)",
    color = "Demographic Group"
  )
```

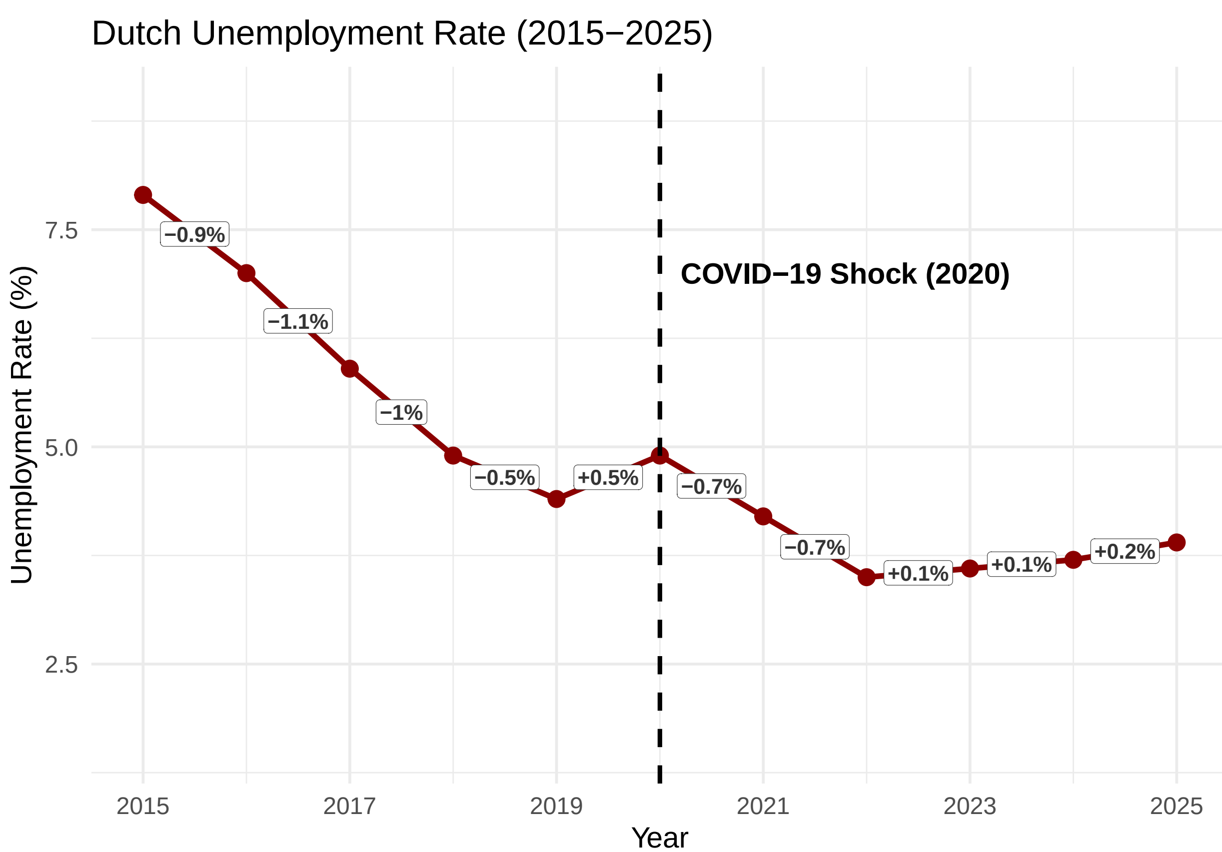


Youth unemployment (ages 15-24) stays consistently far above the total rate across the whole period, peaking near 12.5% in 2015 and spiking again around the 2020 shock before easing. This persistent gap shows that young workers absorb a disproportionate share of labour-market shocks, which is the rationale behind our Youth Vulnerability Ratio.

3.6 Event Analysis

```r
temporal_changes <- combined_data %>%
  arrange(Year) %>%
  mutate(
    Ppt_Change = Unemp_Rate - lag(Unemp_Rate),
    x_mid = (Year + lag(Year)) / 2,
    y_mid = (Unemp_Rate + lag(Unemp_Rate)) / 2,
    Label_Text = ifelse(Ppt_Change > 0,
                        paste0("+", round(Ppt_Change, 1), "%"),
                        paste0(round(Ppt_Change, 1), "%"))
```

```r
  ) %>%
  filter(!is.na(Ppt_Change))

ggplot(data = combined_data, aes(x = Year, y = Unemp_Rate)) +
  geom_line(color = "darkred", linewidth = 1) +
  geom_point(color = "darkred", size = 2.5) +
  geom_vline(xintercept = 2020, linetype = "dashed", color = "black", linewidth = 0.8) +
  annotate("text", x = 2020.2, y = 7, label = "COVID-19 Shock (2020)", hjust = 0, fontface = "bold") +

  geom_label(data = temporal_changes,
             aes(x = x_mid, y = y_mid, label = Label_Text),
             fill = "white", color = "gray20", size = 2.8, fontface = "bold",
             label.padding = unit(0.15, "lines"), label.size = 0.1) +

  scale_x_continuous(breaks = seq(2015, 2025, by = 2)) +
  theme_minimal() +
  labs(title = "Dutch Unemployment Rate (2015-2025)", x = "Year", y = "Unemployment Rate (%)") +
  ylim(1.5, 9)
```



The labelled changes show unemployment fell steadily before 2020 (around -1 ppt per year), then rose by only +0.5 ppt in the COVID-19 year. This increase is remarkably small given the sharp GDP contraction that year, most likely because the Dutch emergency wage-subsidy scheme (NOW) kept workers formally employed, temporarily weakening the usual Okun's Law relationship.

Part 4 - Discussion

4.1 Discuss your findings

Our analysis points to four findings. First, Dutch unemployment fell steadily from roughly 8% in 2015 to below 4% by 2025, interrupted only by a small rise around the 2020 COVID-19 shock. Second, GDP growth and unemployment broadly move in opposite directions, consistent with Okun's Law, but 2020 is a clear outlier: GDP fell sharply while unemployment barely moved, most likely because of the Dutch wage-subsidy scheme. Third, youth unemployment is consistently two to three times the total rate and moves more sharply with the cycle, as captured by our Youth Vulnerability Ratio. Fourth, the 2020 spatial comparison shows the Netherlands among the most resilient labour markets in Western Europe.

These results should be read with caution. We document associations, not causal effects: GDP and unemployment are both shaped by other factors that affect them at the same time, such as labour-market policy, demographic change, and measurement differences across Eurostat series. The 2020 point shows clearly how a simple correlation can mislead. Limitations include the short annual series (2015-2025), the focus on the Netherlands for the time-series analysis, and the use of a single year (2020) for the spatial map.

Part 5 - Reproducibility

5.1 Github repository link

https://github.com/jackievahtera/PFE_Group_Project

5.2 How to reproduce this report

The repository contains this report (`betaversion.Rmd`), a `data/` folder with the three Eurostat CSV extracts (`raw_gdp.csv`, `raw_unemployment_sex.csv`, `raw_unemployment_youth.csv`), the bibliography (`references.bib`), and a `README.md` with a project overview.

To reproduce the report: (1) clone the repository, (2) open `ProgrForEconom.Rproj` in RStudio, (3) make sure the three CSVs are in the `data/` folder, (4) install the required packages (`tidyverse`, `maps`), and (5) knit `betaversion.Rmd` to PDF. Knitting re-reads the raw data and regenerates every figure and statistic in the document. All file paths in the `.Rmd` are relative to the project root, so the project runs on any machine without changes.

5.3 Reference list

Eurostat. (2025). Real GDP growth rate; Total unemployment rate; Youth unemployment rate [Data sets, tables tec00115, tesem120, tipslm80]. https://ec.europa.eu/eurostat

Eurofound. (2017). *Long-term unemployment.* Publications Office of the European Union.

Okun, A. M. (1962). Potential GNP: Its measurement and significance. *Proceedings of the Business and Economic Statistics Section*, 98–104.